

NAPO, China

WMO: 592090

Lat: 23.4142N

Lon: 105.8200E

Elev: 2605

StdP: 13.36

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 38.2	(c) 40.6	(d) 29.1	(e) 25.4	(f) 46.4	(g) 32.0	(h) 29.0	(i) 44.9	(j) 12.3	(k) 59.5	(l) 10.6	(m) 58.8	(n) 2.5	(o) 290	(p) 0.256

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.5	(c) 89.9	(d) 73.4	(e) 88.2	(f) 73.4	(g) 86.5	(h) 72.9	(i) 76.2	(j) 85.0	(k) 75.5	(l) 83.8	(m) 74.8	(n) 82.6	(o) 5.2	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	139.1	79.7	73.3	136.6	79.2	72.7	133.8	78.7	41.9	85.3	41.2	84.3	40.5	83.0	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.0	(b) 8.7	(c) 7.6	(e) 33.6	(f) 93.5	(g) 2.8	(h) 1.3	(i) 31.6	(j) 94.4	(k) 29.9	(l) 95.2	(m) 28.4	(n) 96.0	(o) 26.4	(p) 96.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	68.0	53.3	57.5	63.7	71.1	75.2	77.6	78.1	77.4	74.4	69.3	62.7	55.2
	DBStd	10.46	7.53	8.84	7.33	6.60	5.06	3.40	2.73	2.78	3.96	4.68	5.90	6.12
	HDD50	97	52	26	1	0	0	0	0	0	0	0	1	17
	HDD65	1185	367	237	117	23	1	0	0	0	1	17	112	310
	CDD50	6669	154	237	427	632	781	829	871	850	731	597	382	178
	CDD65	2284	5	28	78	204	318	379	406	385	282	149	43	7
	CDH74	14092	7	104	325	1207	2224	2662	2793	2466	1618	565	112	9
	CDH80	3726	0	8	55	335	642	685	762	754	414	67	4	0
Wind	WSAvg	3.4	3.7	4.0	4.3	4.3	3.8	3.3	3.0	2.6	2.7	3.0	3.2	3.2
Precipitation	PrecAvg	55.2	0.7	0.7	1.4	2.6	6.5	9.6	10.2	11.1	6.2	3.6	2.0	0.7
	PrecMax	73.8	1.6	2.3	6.7	5.0	15.2	16.4	20.2	23.3	11.5	8.7	5.4	4.3
	PrecMin	39.2	0.0	0.0	0.1	0.3	1.6	2.6	4.8	4.0	2.5	0.7	0.2	0.0
	PrecStd	8.5	0.4	0.6	1.6	1.4	2.7	3.8	3.9	4.6	2.2	1.9	1.6	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.8	81.2	84.9	90.5	92.1	90.3	90.9	91.4	89.2	84.3	80.4	75.1
		MCWB	61.1	62.4	66.0	69.7	72.8	74.8	74.8	74.3	73.2	69.2	67.4	63.4
	2%	DB	70.6	77.3	81.1	87.2	88.9	88.1	88.6	88.8	86.7	81.9	77.3	71.1
		MCWB	60.3	62.7	66.4	70.2	72.7	74.4	74.4	74.0	72.0	69.0	65.4	61.1
	5%	DB	67.4	73.7	77.7	83.9	86.1	86.2	86.6	86.7	84.7	79.7	74.7	68.0
		MCWB	59.3	62.3	65.8	70.0	71.7	74.0	74.3	73.6	71.5	68.1	64.1	59.3
	10%	DB	64.3	69.9	74.1	80.9	83.5	84.1	84.4	84.5	82.3	77.3	72.2	64.7
		MCWB	58.3	61.5	65.1	69.5	71.3	73.7	73.9	73.3	71.1	67.5	63.7	57.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.4	65.7	69.3	73.5	75.8	77.0	77.2	77.0	75.5	72.8	69.4	65.1
		MCDB	69.2	75.5	79.2	84.0	86.2	85.9	86.6	86.5	84.2	78.8	76.2	72.2
	2%	WB	61.4	64.3	67.8	72.1	74.7	76.1	76.1	75.7	74.3	71.3	68.1	62.8
		MCDB	68.1	72.7	76.9	81.8	83.9	84.4	84.7	84.3	82.0	77.6	73.8	69.2
	5%	WB	60.1	63.2	66.7	71.1	73.7	75.3	75.3	75.0	73.4	70.3	66.7	60.9
		MCDB	66.3	71.0	75.3	80.1	82.1	83.1	83.2	83.0	80.3	76.1	72.3	66.2
	10%	WB	58.7	61.9	65.4	70.2	72.9	74.6	74.6	74.2	72.6	69.3	65.4	59.0
		MCDB	64.3	69.2	73.2	78.6	80.7	81.7	81.8	81.5	78.9	74.9	70.6	64.0
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.7	11.9	12.5	12.6	11.3	11.5	13.1	13.8	12.2	13.2	12.1
		MCDBR	15.0	17.4	17.0	16.8	16.7	14.6	15.6	17.7	17.9	16.9	17.1	17.1
		MCWBR	6.7	6.0	5.9	5.0	4.6	4.3	4.4	5.2	5.2	5.7	6.7	7.5
	5% WB	MCDBR	12.6	14.7	14.9	14.3	13.0	11.7	13.1	14.8	13.3	12.0	11.3	13.1
		MCWBR	6.0	5.9	6.2	5.1	4.5	4.2	4.4	5.1	4.5	4.7	4.9	6.4
Clear-Sky Solar Irradiance	taub	0.521	0.528	0.683	0.645	0.504	0.499	0.476	0.530	0.549	0.610	0.530	0.556	
	taud	1.836	1.859	1.554	1.690	2.116	2.157	2.211	2.035	1.948	1.735	1.864	1.752	
	Ebn at Noon	230	240	210	223	256	255	261	247	240	215	225	212	
	Edh at Noon	60	62	89	78	51	49	46	55	59	69	58	63	
All-Sky Solar Radiation	RadAvg	714	975	1126	1394	1469	1374	1459	1428	1318	1016	970	752	
	RadStd	145	185	105	135	159	139	95	91	99	106	126	97	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.95	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.50	N/A	N/A	+1.01	N/A	N/A	N/A	N/A	N/A	+133

Nomenclature: See separate page